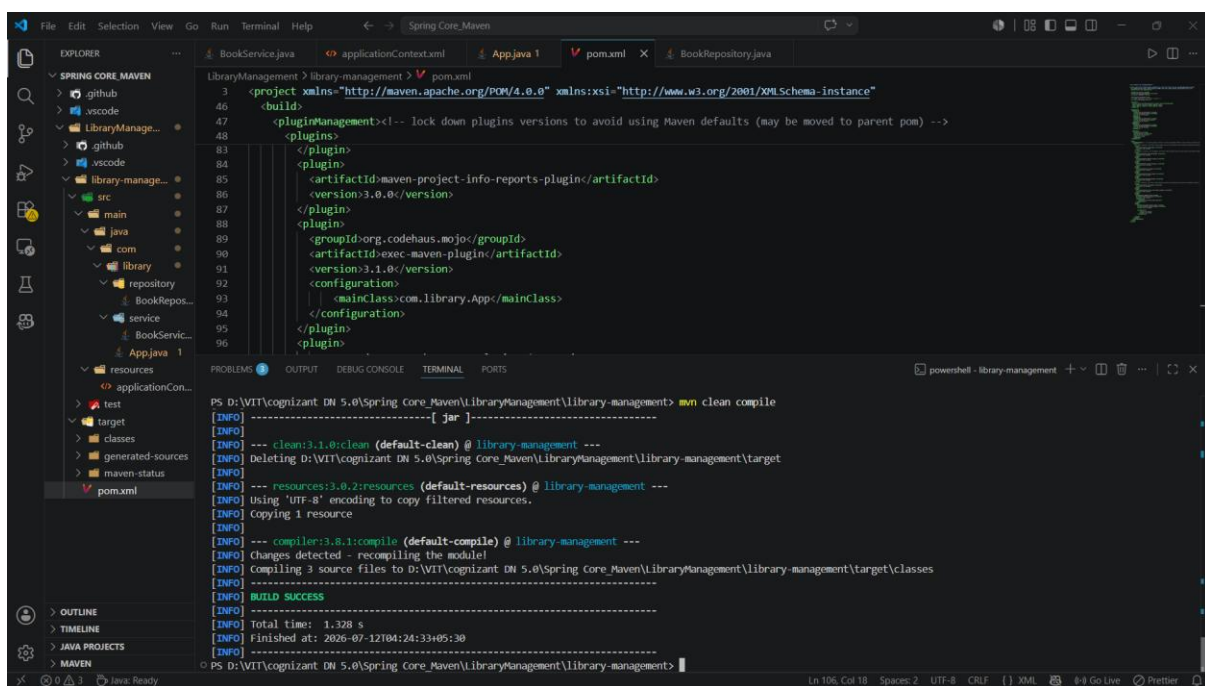


Spring Core Maven Exercise 2 | Cognizant DN 5.0

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Exercise 4 - Creating and Configuring a Maven Project

1. Created a Maven project named LibraryManagement.
2. Configured the project using the pom.xml file.
3. Added the Spring Context dependency.
4. Added the Spring AOP dependency.
5. Added the Spring WebMVC dependency.
6. Configured the Maven Compiler Plugin with Java version 1.8.
7. Downloaded the required Maven dependencies.
8. Built the project using Maven.
9. Verified successful compilation of the project.



The screenshot shows an IDE with the following components:

- EXPLORER:** Displays the project structure for 'Spring Core Maven'. The 'pom.xml' file is selected under the 'resources' folder.
- Editor:** Shows the content of 'pom.xml' with line numbers 3 to 96. The XML content is as follows:

```
3 <?xml version="1.0" encoding="UTF-8"?>
4 <!-- Maven Project -->
5 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
6         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
7     <modelVersion>4.0.0</modelVersion>
8     <groupId>com.library</groupId>
9     <artifactId>library-management</artifactId>
10    <version>1.0.0</version>
11    <packaging>jar</packaging>
12    <name>Library Management</name>
13    <description>Library Management</description>
14    <url>http://maven.apache.org</url>
15    <inceptionYear>2016</inceptionYear>
16    <licenses>
17        <license>
18            <name>The Apache Software License, Version 2.0</name>
19            <url>http://www.apache.org/licenses/LICENSE-2.0.txt</url>
20        </license>
21    </licenses>
22    <developers>
23        <developer>
24            <id>sanskarkumawat</id>
25            <name>SANSKAR KUMAWAT</name>
26            <email>sanskarkumawat@gmail.com</email>
27        </developer>
28    </developers>
29    <dependencies>
30        <dependency>
31            <groupId>org.springframework</groupId>
32            <artifactId>spring-context</artifactId>
33            <version>5.0.0</version>
34            <scope>compile</scope>
35        </dependency>
36        <dependency>
37            <groupId>org.springframework</groupId>
38            <artifactId>spring-aop</artifactId>
39            <version>5.0.0</version>
40            <scope>compile</scope>
41        </dependency>
42        <dependency>
43            <groupId>org.springframework</groupId>
44            <artifactId>spring-webmvc</artifactId>
45            <version>5.0.0</version>
46            <scope>compile</scope>
47        </dependency>
48    </dependencies>
49    <build>
50        <pluginManagement><!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
51            <plugins>
52                <plugin>
53                    <groupId>org.apache.maven.plugins</groupId>
54                    <artifactId>maven-project-info-reports-plugin</artifactId>
55                    <version>3.0.0</version>
56                </plugin>
57                <plugin>
58                    <groupId>org.codehaus.mojo</groupId>
59                    <artifactId>exec-maven-plugin</artifactId>
60                    <version>3.1.0</version>
61                    <configuration>
62                        <mainClass>com.library.App</mainClass>
63                    </configuration>
64                </plugin>
65            </plugins>
66        </pluginManagement>
67        <plugins>
68            <plugin>
69                <groupId>org.apache.maven.plugins</groupId>
70                <artifactId>maven-compiler-plugin</artifactId>
71                <version>3.8.1</version>
72                <configuration>
73                    <source>1.8</source>
74                    <target>1.8</target>
75                </configuration>
76            </plugin>
77        </plugins>
78    </build>
79    <properties>
80        <java.version>1.8</java.version>
81    </properties>
82    <test>
83        <resources>
84            <resource>
85                <directory>src/test/resources</directory>
86            </resource>
87        </resources>
88        <classes>
89            <directory>src/test/classes</directory>
90        </classes>
91    </test>
92    <profiles>
93        <profile>
94            <id>dev</id>
95            <activation>
96                <activeByDefault>true</activeByDefault>
97            </activation>
98            <properties>
99                <env.dir>D:\VIT\cognizant DN 5.0\Spring Core Maven\LibraryManagement\library-management</env.dir>
100            </properties>
101        </profile>
102    </profiles>
103</project>
```
- Terminal:** Shows the output of the command 'mvn clean compile'. The output indicates that the project was successfully built, with the following details:

```
PS D:\VIT\cognizant DN 5.0\Spring Core Maven\LibraryManagement\library-management> mvn clean compile
[INFO] -----[ jar ]-----
[INFO] --- clean:3.1.0:clean (default-clean) @ library-management ---
[INFO] Deleting D:\VIT\cognizant DN 5.0\Spring Core Maven\LibraryManagement\library-management\target
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO] --- compiler:3.8.1:compile (default-compile) @ library-management ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to D:\VIT\cognizant DN 5.0\Spring Core Maven\LibraryManagement\library-management\target\classes
[INFO] BUILD SUCCESS
[INFO] Total time: 1.328 s
[INFO] Finished at: 2026-07-12T04:24:33+05:30
[INFO] -----
```

The screenshot shows an IDE window with the title "Spring Core Maven". The interface is divided into several panes:

- EXPLORER:** Shows a project structure with folders like "src", "main", "java", "com", "library", "repository", "service", "resources", "test", "classes", "generated-sources", "maven-status", and "pom.xml".
- PROBLEMS:** Empty.
- OUTPUT:** Displays the Maven dependency tree and build output. The dependency tree lists various artifacts being downloaded from central repositories, including commons-io, commons-codec, commons-compress, plexus-io, plexus-utils, maven-dependency-analyzer, maven-artifact-transfer, and various Spring Framework and JUnit dependencies. The build output shows the execution of the Maven build process, including the compilation of source files and the successful completion of the build.
- TERMINAL:** Shows the command prompt with the command "mvn dependency:tree" and its output.

The terminal output shows the following commands and results:

```
PS D:\VIT\cognizant DN 5.0\Spring Core Maven\Library\management\library-management> mvn dependency:tree
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-11/zstd-jni-1.5.5-11.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.2/plexus-io-3.4.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.26.1/commons-compress-1.26.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-i18n/1.0-beta-10/plexus-i18n-1.0-beta-10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-analyzer/1.14.1/maven-dependency-analyzer-1.14.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-tree/3.3.0/maven-dependency-tree-3.3.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-i18n/1.0-beta-10/plexus-i18n-1.0-beta-10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-artifact-transfer/0.13.1/maven-artifact-transfer-0.13.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-analyzer/1.14.1/maven-dependency-analyzer-1.14.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-component-annotations/2.0.0/plexus-component-annotations-2.0.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.3.2/maven-common-artifact-filters-3.3.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-artifact-transfer/0.13.1/maven-artifact-transfer-0.13.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-component-annotations/2.0.0/plexus-component-annotations-2.0.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-11/zstd-jni-1.5.5-11.jar
[INFO] com.library:library-management:jar:1.0-SNAPSHOT
[INFO] +- org.springframework:spring-context:jar:5.3.30:compile
[INFO] | +- org.springframework:spring-beans:jar:5.3.30:compile
[INFO] | +- org.springframework:spring-core:jar:5.3.30:compile
[INFO] | | \- org.springframework:spring-jcl:jar:5.3.30:compile
[INFO] | \- org.springframework:spring-expression:jar:5.3.30:compile
[INFO] +- org.springframework:spring-aop:jar:5.3.30:compile
[INFO] +- org.springframework:spring-webmvc:jar:5.3.30:compile
[INFO] | \- org.springframework:spring-web:jar:5.3.30:compile
[INFO] \- junit:junit:jar:4.13.2:test
[INFO] \- org.hamcrest:hamcrest-core:jar:1.3:test
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.487 s
[INFO] Finished at: 2026-07-12T04:25:04+05:30
[INFO] -----
```